1. [15 minutes] You have a machine that can only perform multiplication and addition on **single-digit** numbers.

You want to use Karatsuba's algorithm to multiply two-digit numbers.

Karatsuba's formula for multiplying two large integer numbers is:

$$(x \times 10^m + y) \times (z \times 10^m + t) = p \times 10^{2m} + (r - p - q) \times 10^m + q$$

$$p = x \times z, \qquad q = y \times t, \qquad r = (x + y) \times (z + t)$$

a) How many single-digit multiplications are needed to compute **a = 24** × **35**? Explain why. (5 points)

b) How many single-digit multiplications are needed to compute **b = 56** × **78**? Explain why. (5 points)

a) 3

$\boldsymbol{p = 2 \times 3, q = 4 \times 5, r = 6 \times 8}$

b) 5

$\boldsymbol{p = 5 \times 7, q = 6 \times 8, r = 11 \times 15 \rightarrow p = 1 \times 1, q = 1 \times 5, r = 2 \times 6}$: $\boldsymbol{3\ multiplcatins}$

2+3=5

2 + 3 = 5